

|                |                    |
|----------------|--------------------|
| Course Number  | COE 328            |
| Course Title   | Digital Systems    |
| Semester/Year  | Semester 3, Year 2 |
| Section Number | 1                  |

## Lab 4 - Circuit & Storage Elements

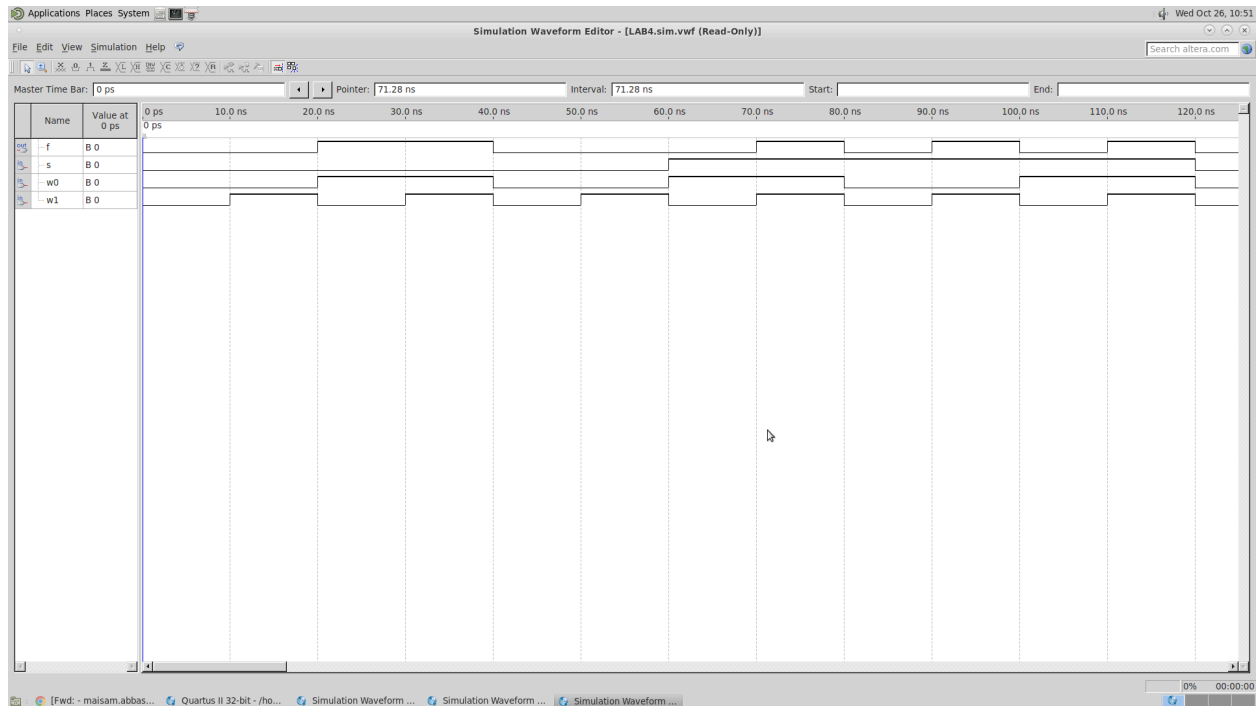
|                  |                     |
|------------------|---------------------|
| Assignment Title | Lab 4               |
| Submission Date  | th, 2022            |
| Due Date         | November 11th, 2022 |

| Student Name      | Student ID (xxxx1234) | Signature* |
|-------------------|-----------------------|------------|
| Ahmad Najafi      | 24585                 | Najafi     |
| Syed Maisam Abbas | 03255                 | Maisam     |

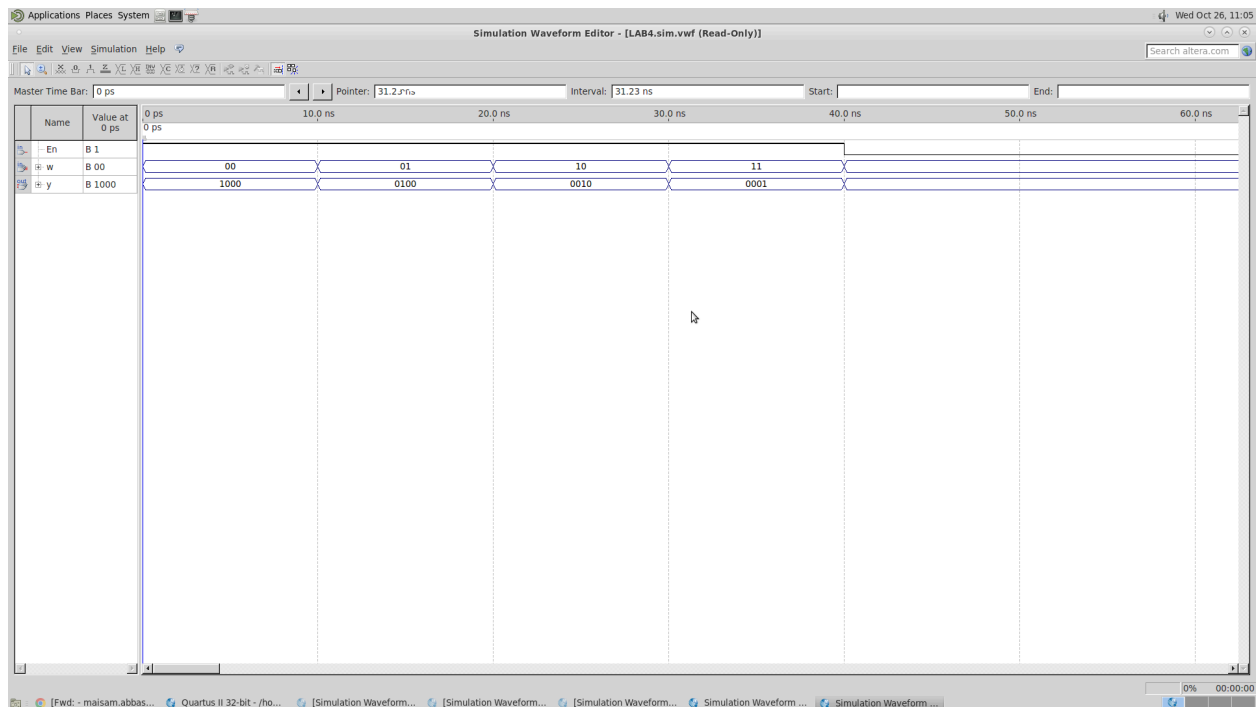
(Note: Remove the first 4 digits from your student ID)

*\*By signing above you attest that you have contributed to this submission and confirm that all work you have contributed to this submission is your own work. Any suspicion of copying or plagiarism in this work will result in an investigation of Academic Misconduct and may result in a "0" on the work, an "F" in the course, or possibly more severe penalties, as well as a Disciplinary Notice on your academic record under the Student Code of Academic Conduct, which can be found online at: <http://www.ryerson.ca/senate/policies/pol60.pdf>.*

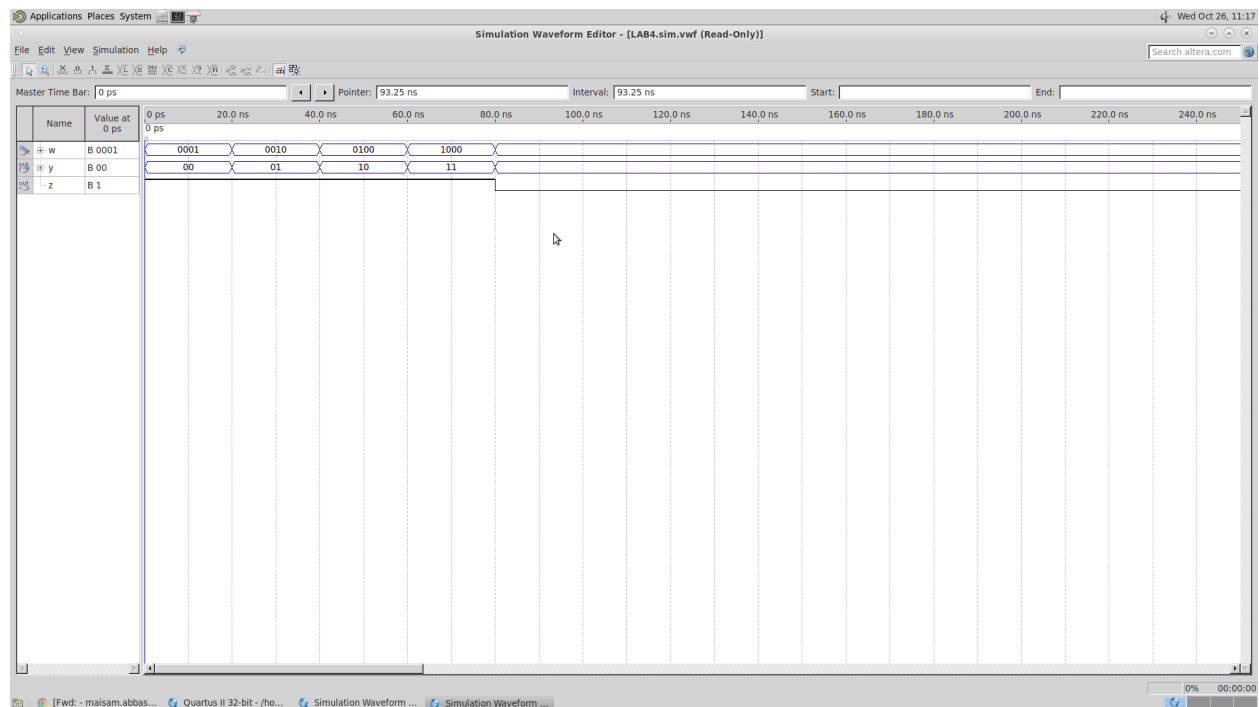
## In-Lab:



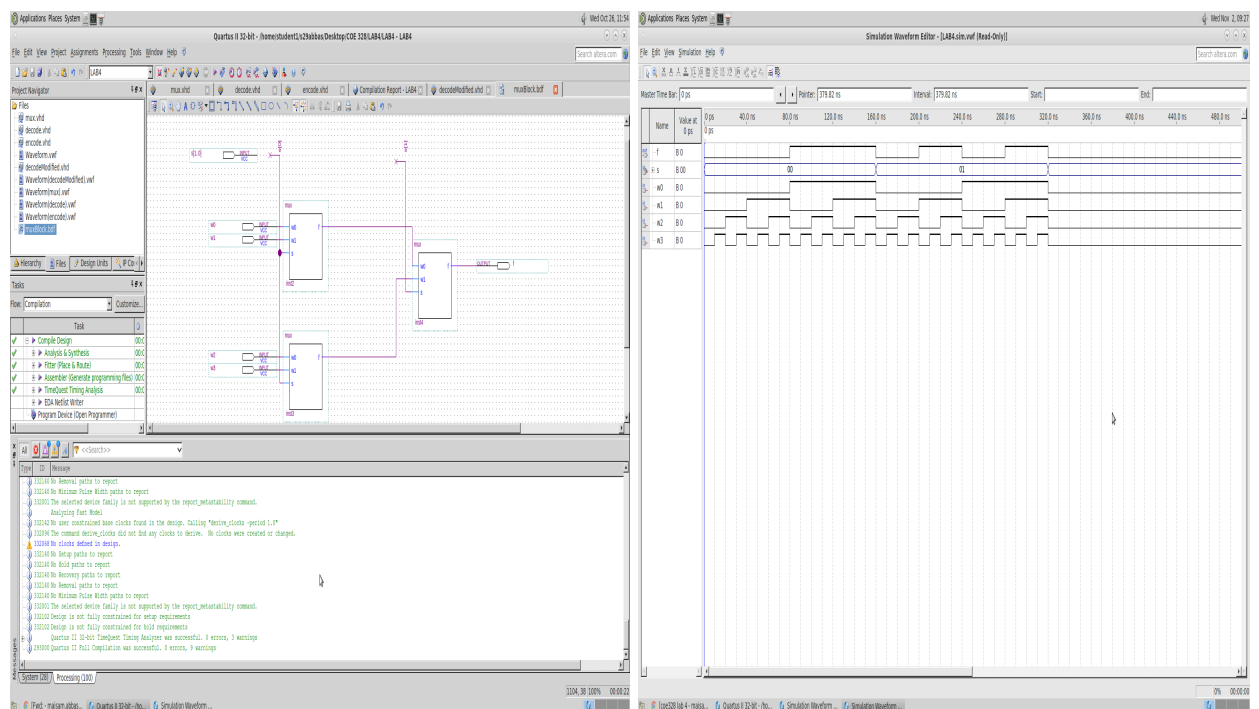
**Figure 1:** Waveform of 2-to-1 multiplexer



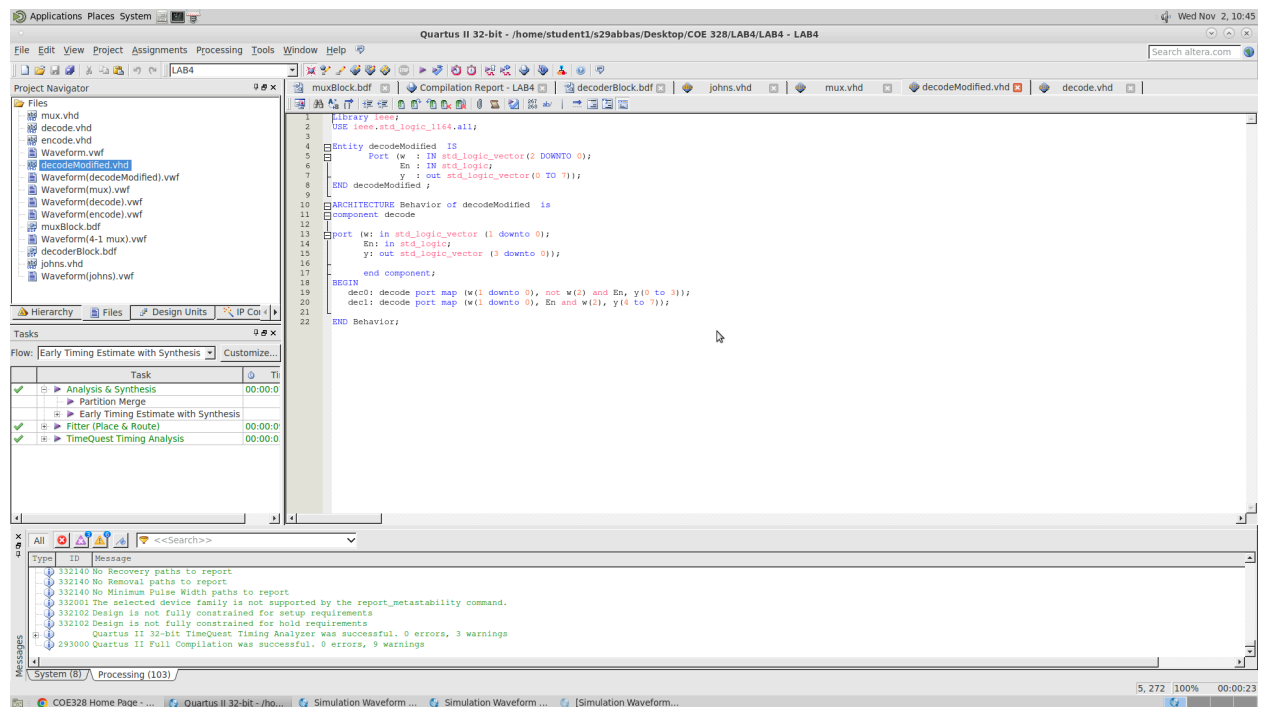
**Figure 2:** Waveform of 2:4 decoder



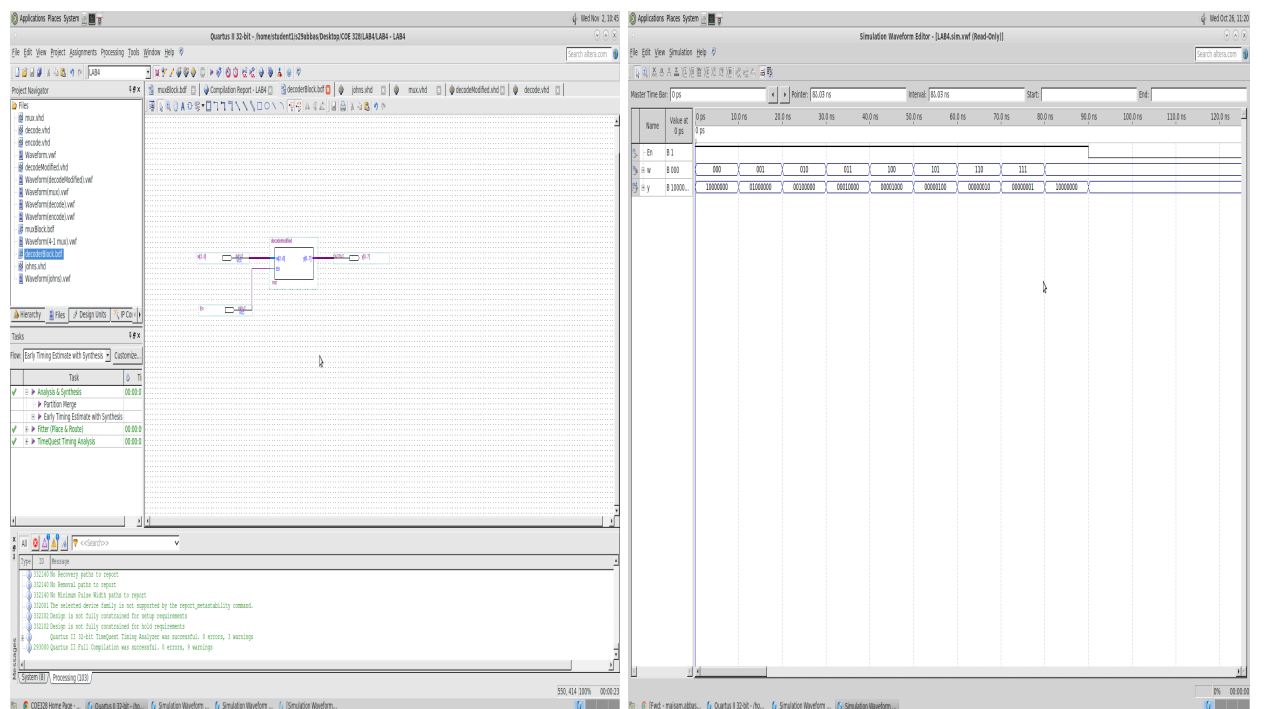
**Figure 3: Waveform of priority encoder**



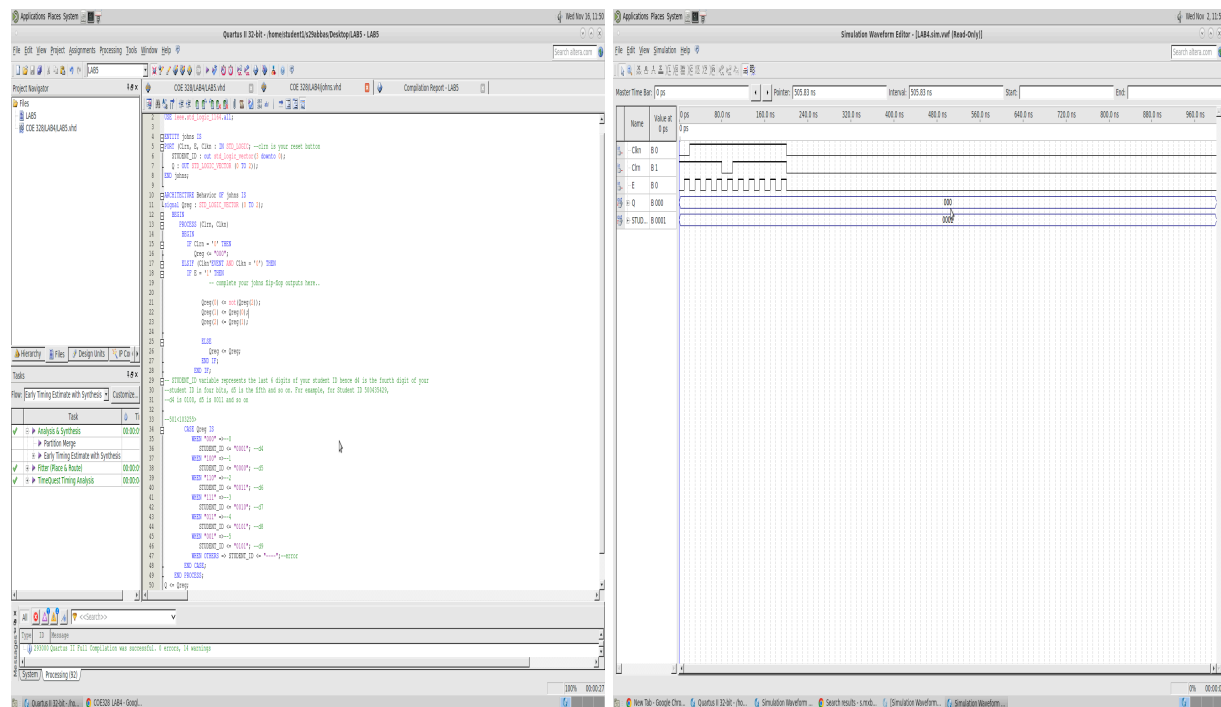
**Figure 4 & 5: Block diagram and waveform of 4-to-1 multiplexer**



**Figure 6:** Modified VHDL code for decoder (2:4 to 3:8)



**Figure 7 & 8:** Block diagram and waveform of 3:8 decoder



**Figure 9 & 10: VHDL code and waveform for Johnson Counter**

Using the VHDL code, the case code ensures that the 3-bit binary is converted to a 4-bit binary number to configure the required student number. Additionally, the output of the proceeding flip-flop is connected as the input of the next flip-flop. Therefore, this waveform represents the given student number.

|   | Q0 | Q1 | Q2 |
|---|----|----|----|
| 0 | 0  | 0  | 0  |
| 1 | 1  | 0  | 0  |
| 2 | 1  | 1  | 0  |
| 3 | 1  | 1  | 1  |
| 4 | 0  | 1  | 1  |
| 5 | 0  | 0  | 1  |

**Table 1: John's truth table**

Student ID: 501103255

0101 0000 0001 0001 0000 0011 0010 0101 0101  
 1 0 3 2 5 5

last 6 digits of Student number

| $x_3 x_4$ | 00 | 01 | 11 | 10 |
|-----------|----|----|----|----|
| $x_1 x_2$ | 00 | 1  | 1  | 1  |
|           | 01 |    |    |    |
|           | 11 |    |    |    |
|           | 10 |    |    |    |

$$f(x_1, x_2, x_3, x_4) = m\sum(1, 0, 3, 2, 5)$$

$$f = \bar{x}_1 \bar{x}_2 + \bar{x}_1 \bar{x}_3 x_4$$

$$g = \bar{x}_1 \bar{x}_2, \quad h = \bar{x}_1 \bar{x}_3 x_4$$

• Truth table

| m  | $x_1$ | $x_2$ | $x_3$ | $x_4$ | g | h | f |
|----|-------|-------|-------|-------|---|---|---|
| 0  | 0     | 0     | 0     | 0     | 1 | 0 | 1 |
| 1  | 0     | 0     | 0     | 1     | 1 | 1 | 1 |
| 2  | 0     | 0     | 1     | 0     | 1 | 0 | 1 |
| 3  | 0     | 0     | 1     | 1     | 1 | 1 | 1 |
| 4  | 0     | 1     | 0     | 0     | 0 | 0 | 0 |
| 5  | 0     | 1     | 0     | 1     | 0 | 0 | 0 |
| 6  | 0     | 1     | 1     | 0     | 0 | 0 | 0 |
| 7  | 0     | 1     | 1     | 1     | 0 | 0 | 0 |
| 8  | 1     | 0     | 0     | 0     | 0 | 0 | 0 |
| 9  | 1     | 0     | 0     | 1     | 0 | 0 | 0 |
| 10 | 1     | 0     | 1     | 0     | 0 | 0 | 0 |
| 11 | 1     | 0     | 1     | 1     | 0 | 0 | 0 |
| 12 | 1     | 1     | 0     | 0     | 0 | 0 | 0 |
| 13 | 1     | 1     | 0     | 1     | 0 | 0 | 0 |
| 14 | 1     | 1     | 1     | 0     | 0 | 0 | 0 |
| 15 | 1     | 1     | 1     | 1     | 0 | 0 | 0 |

$$g = \bar{x}_1 \bar{x}_2$$

$$h = \bar{x}_1 \bar{x}_3 x_4$$

$$f = \bar{x}_1 \bar{x}_2 + \bar{x}_1 \bar{x}_3 x_4$$

Figure 11: K-map and truth table for student ID